Wholesale Service Management v8.5.0

Vocus Wholesale Web Services

Commercial and in confidence

Vocus Group Limited | ABN 96 084 115 499 | vocus.com.au
Level 10, 452 Flinders Street, Melbourne
Victoria 3000, AUSTRALIA

Copyright Notice

This document is provided by Vocus Group Limited on the basis that you will treat it as confidential. No part of this documentation may be reproduced or copied in any form by any means without the written permission of Vocus Group Limited. Unless otherwise expressly agreed in writing, the information contained in this document is subject to change without notice and Vocus Group Limited assumes no responsibility for any alteration to, or any error or other deficiency, in this document.

Copyright © 2009 - 2025 Vocus Group Limited. All rights reserved.

1 Table of Contents

1 Table of Contents 1
2 Overview 2
2.1 WSM Resources 2
2.2 Contacting Vocus 2
3 Accessing WSM 3
3.1 Access Application Form 3
3.2 Test Environment Access 3
3.3 Production Environment Access 3
4 WSM Authentication & Restrictions 4
5 SOAP Fault Management 5
5.1 Basic Error Codes 5
5.2 WSM Soap Fault Format 5
6 Web Service Operations 6
6.1 Operation WSM::Create 6
6.2 Operation WSM::Get 7
6.3 Operation WSM::Set 7
6.4 Operation WSM::Delete 7
7 Web Service Call-Back & Notifications 9
7.1.1 Call Back Authentication 9
7.1.2 Call-Back SOAP Server Specification 9
7.1.3 Call Back Message Structure 10
7.1.4 SOAP Fault Management 10
8 Test Environment 11
8.1 Accessing the Sandbox 11
8.2 Simulating Successful Operation 11
8.3 Simulating Failed Operation 11
8.4 Simulating WSM's Generated Transaction 11

2 Overview

Wholesale Service Management Web Service (WSM) provides Vocus clients with the ability to perform common provisioning, runtime and reporting operations on many of Vocus services via an electronic B2B interface. This is an alternative to the client's staff using one of Vocus Web GUI interfaces to perform the same operations manually.

WSM is based on SOAP 1.1 running on a firewall protected HTTPS 1.1 POST protocol. Most WSM services operate in an asynchronous manner, using (optional) SOAP call-backs to the client's network to provide notification of transaction results. All data types are standardised.

WSM operates in **AEST** as the standard time zone as defined by the NTP timeserver **au.pool.ntp.org.**

2.1 WSM Resources

The resources provided to the client as part of the WSM release are:

- Vocus Wholesale Service Management Description Document (this document)
- Vocus Wholesale Service Management Examples
- Product Catalogue Document for TCAS (Transaction Control and Status Service)
- Product Catalogue Document for DIR (Directory Service)
- Commercial Product Catalogue Documents, together with supporting documents if applicable
- Vocus Standard Data Types Definition which applies to all documents in this SDK
- Vocus Street Address Component and Post Code Standard in Excel format
- WholesaleServiceManagement.wsdl
- WholesaleServiceManagement.xsd
- StandardDataTypes.xsd
- WSDL template for SOAP call-backs

2.2 Contacting Vocus

All requests for support relating to WSM should be lodged to the Vocus Wholesale Service Assurance. To avoid delays, any fault tickets lodged should include the SOAP **TransactionID** and any other details that will help Vocus to identify the fault.

3 Accessing WSM

3.1 Access Application Form

For all WSM access request, please use **Access Request.docx** and forward it to Vocus Wholesale Service Assurance team.

3.2 Test Environment Access

To enable the client to access the test environment, Vocus Wholesale will provide the following components:

- Authentication certificate in keystore format + password
- Maximum of 3 x WSM AccessKeys
- Vocus IP address as call-back source
- Vocus AccessKey for call-back authentication

In addition Vocus Wholesale requires the following information from the client:

- The client web service client static IP address. Vocus will allow access for up to 2 IP addresses
- Client SOAP call-back destination hostname & IP address
- WholesaleServiceManagementCallBack.wsdl

For further details about the WSM test environment, please see Chapter 8, Test Environment.

3.3 Production Environment Access

To gain access to production environment, client must first perform testing against the sandbox environment. Once this is completed, client will need to notify Vocus Wholesale to review the SOAP request sent to the testing environment.

For example, to gain access to production system for NBN provisioning, the client will need to demonstrate the understanding of WSM flows for NBN provisioning. This will consist of but not limited to:

- Understanding the login process
- Performing service qualification
- Sending the correct request for NBN provisioning

Vocus will only grant access to the SOAP method client has tested at the sandbox environment.

Access to the production environment requires Vocus Wholesale and the client to each provide the equivalent production system components as were provided for the test environment.

4 WSM Authentication & Restrictions

The client's web services application will login using HTTP Client Certificate Authentication. This is also known as two way SSL.

The URL for certificate authentication is: **https://wsm.webservice.m2.com.au:9443/login/**

The certificate login sequence is as follow:

1. Client sends the certificate to WSM
2. WSM verifies the certificate
3. For a valid certificate, WSM will reply with HTTP Status Code 202 (Accepted)
4. Replies containing values other than HTTP Status Code 202 indicate that WSM has rejected the certificate.

Following a successful certificate authentication request, WSM will create a transaction session that has an idle timeout of 600 seconds. The client will then send SOAP requests for that session to the URL specified in the WSDL file.

When executing an operation a special identifier **AccessKey** is sent with the SOAP request. One login certificate can be associated with 10 client AccessKeys.

Vocus will provide a new certificate to the client on an annual basis. There will be a 3 month grace period following the issue of the new certificate before the existing certificate is revoked. This will provide the client adequate time to replace the previous certificate with the newly issued certificate.

Vocus imposes limits on the rate at which the client SOAP application may call web service functions. This is done in order to protect the service from degradation and to prevent flooding / DDoS attacks. If the limit is exceeded the client is restricted from making any further requests for a short period. The default restricted period is 10 seconds. The default rate limit is 5 requests per 10 seconds which is applied at the overall web service level.

5 SOAP Fault Management

WSM will return **SOAP faults** with a *faultstring, ErrorCode*, and *TransactionID*. Client is advised to utilise *ErrorCode* for detecting error, *faultstring* is less reliable as the message can change anytime.

Some faults will not have an ErrorCode due to the request failing before it reaches the WSM processing layers. For this type of error, generally a human intervention is required.

SOAP faults can be returned synchronously or asynchronously. Synchronous operation can cause synchronous faults, while asynchronous operation can cause both synchronous and asynchronous faults depending on where the error is captured.

5.1 Basic Error Codes

The following table lists all basic error codes. Product specific error messages are available in the Product Catalogue documentation for each product.

ErrorCode	faultstring	Description
WSM-0000	Internal error	Operation failed because of internal error
WSM-4000	General client error	Default client error code
WSM-4005	Login required	Client need to authenticate
WSM-4020	Account blocked. Contact Vocus	WSM account is blocked. Contact Vocus
WSM-4050	Insufficient access privileges	Trying to access a product or operation that the client is not authorised to access
WSM-4080	Certificate request rate limit exceeded	Client has exhausted the allocated operation quota at the certificate level
WSM-4220	Various messages, depending on what the invalid XML are	XML validation error
WSM-4221	Invalid ProductID	Client is trying to access an invalid product
WSM-4224	Invalid Operation	Client tries to execute an invalid operation
WSM-5000	General server error	Default server error code
WSM-5010	Partially completed	Not all backend and syndicated operation is completed. If possible use ProductID TCAS to query the transaction. Open a support ticket to Vocus.
WSM-5030	Syndicated service error	One of the supporting systems has experienced an error while processing the request. Open a support ticket to Vocus
WSM-5050	Not implemented	The operation is not yet implemented

5.2 WSM Soap Fault Format

Synchronous faults will use the following SOAP fault format. Asynchronous faults will use the fault format as defined in ProductID TCAS. For more details on the mechanism, refer to the examples chapter.

```
<SOAP-ENV:Fault>
   <faultcode>SOAP-ENV</faultcode>
   <faultstring>faultstring<faultstring>
   <detail>
      <ns1:Fault>
         <TransactionID>TransactionID</TransactionID>
         <ErrorCode>ErrorCode</ErrorCode>
      <ns1:Fault>
    </detail>
<SOAP-ENV:/Fault>
```

6 Web Service Operations

WSM provides four different web service operations: Create, Get, Set and Delete.

All WSM web service operations provide a generic interface and must be used in conjunction with the specific values and information described in the individual WSM ***Product Catalogues***. One or more of the following elements are required for each operation:

- ***ProductID*** – the product to manipulate
- ***PlanID*** – the selected plan relating to the product. A single product can have multiple plans.
- ***Scope*** – the provisioning or operational area specific to the request.
- ***Parameters*** – a compound element consisting of one or more Params, a key value pair.

The ProductID and PlanID are also used to provide access to WSM functionality that would otherwise require separate interfaces. For example the Transaction Control and Status (TCAS) "product" is not a product in the sense of a commercial end-user product such as an NBN Service but is nonetheless essential functionality in the management of WSM asynchronous requests. One function of TCAS is to retrieve the result of an asynchronous transaction request via polling instead of relying on the SOAP call-back mechanism.

Unless otherwise specified, all WSM operations are case sensitive and asynchronous. Refer to the Examples chapter for examples on how to build the SOAP messages.

When executing an *asynchronous* operation, WSM returns only the TransactionID. Any operation return parameters will be returned asynchronously via a SOAP call back or by polling TCAS. In *synchronous* mode, WSM will immediately return the SOAP response's TransactionID **AND** parameters**.**

6.1 Operation WSM::Create

Create a new record and provision to Vocus and / or upstream supplier's database and operational infrastructure.

SOAP Request

Element	Cardinality	Notes
AccessKey	1..1	
ProductID	1..1	
PlanID	0..1	
Scope	0..1	
Parameters	0..1	

SOAP Response

Element	Cardinality	Notes
TransactionID	1..1	
ResponseType	1..1	
Parameters	0..1	

6.2 Operation WSM::Get

This operation retrieves details of an existing record.

SOAP Request

Element	Cardinality	Notes
AccessKey	1..1	
ProductID	1..1	
PlanID	0..1	
Scope	0..1	
Parameters	0..1	

SOAP Response

Element	Cardinality	Notes
TransactionID	1..1	
ResponseType	1..1	
Parameters	0..1	

6.3 Operation WSM::Set

This operation allows the client to perform manipulation of an existing record.

Element	Cardinality	Notes
AccessKey	1..1	
ProductID	1..1	Not updateable. For lookup only
PlanID	0..1	
Scope	0..1	
Parameters	0..1	Data to be modified

SOAP Response

Element	Cardinality	Notes
TransactionID	1..1	
ResponseType	1..1	
Parameters	0..1	

6.4 Operation WSM::Delete

This operation removes all traces of a record. It should only be used to remove incorrect and unused records. No historical data is maintained for deleted services.

SOAP Request

Element	Cardinality	Notes
AccessKey	1..1	
ProductID	1..1	
PlanID	0..1	
Scope	0..1	
Parameters	0..1	

SOAP Response

Element	Cardinality	Notes
TransactionID	1..1	
ResponseType	1..1	
Parameters	0..1	

7 Web Service Call-Back & Notifications

The SOAP call-back & notification service. WSM API support both call-back and notification patterns.

Both call-back and notification service implement operation CallBack from WholesaleService ManagementCallBack.wsdl. Client will need to implement and host this service at client side.

Call-back are updates for transaction that originally initiated by client. All ProductID with ASYNC API response will support call-back. This service is the alternative from pooling WSM API for updates, instead WSM API will send updates by calling an API at client side.

Notifications are API updates for Vocus initiated events such as NBN outage notification. Notifications are only supported by certain ProductID. These notifications will have “EVENT” as status param. Please refer to our example and ProductID document for further details.

The updates provided by this service will contain a list of TransactionIDs and their types. Further drill down of the TransactionID is available through **ProductID TCAS** operation Get with scope RESPONSE.

Each call back will report a maximum of 100 TransactionIDs. Only transactions that have changed status since the previous call-back or that has been newly initiated by WSM will be included in the report.

A maximum of 10,000 TransactionID will be stored in the call back publishing queue before WSM will start dropping the oldest TransactionIDs. These TransactionIDs are still available for query through ProductID TCAS.

7.1.1 Call Back Authentication

Operation CallBack will use an AccessKey for authentication. The AccessKey will be chosen by Vocus. This credential will be embedded inside the SOAP request.

7.1.2 Call-Back SOAP Server Specification

The client’s SOAP server requires the following characteristics:

- Implement WholesaleServiceManagementCallBack.wsdl
- SOAP 1.1 using HTTPS 1.1 POST protocol

In addition, Vocus Wholesale suggests the following configuration to ensure security is maintained:

- The SOAP server should run on HTTP port 9443
- The SSL certificate should be signed by a valid Certificate Authority
- The SOAP server is protected by a firewall and only allows access from Vocus Wholesale’s provided IP address
- The SOAP server should also verify that all call-backs are received from the Vocus Wholesale IP address

7.1.3 Call Back Message Structure

SOAP Request

Element	Cardinality	Notes
AccessKey	1..1	

Parameters	1..100	Parameters will contain a list of TransactionIDs and their new status. There are 5 possible params (states):
		• QUEUED
		• PROCESSING
		• SUCCESS
		• FAILED
		• WITHDRAWN
		• EVENT
		
		`<Parameters>`
		`  <Param id="QUEUED">C3482BD3677</Param>`
		`  <Param id="PROCESSING">A84CAF32B53</Param>`
		`  <Param id="SUCCESS">B3B791C3677</Param>`
		`  <Param id="FAILED">A71CAF32772</Param>`
		`  <Param id="WITHDRAWN">C83CAF32891</Param>`
		`  <Param id="EVENT">C83CAF39999</Param>`
		`</Parameters>`
		
		Up to 100 TransactionID will be returned in each call back.

SOAP Response

Element	Cardinality	Notes
TransactionID	1..1	Client-supplied TransactionID for the call-back request.

7.1.4 SOAP Fault Management

During any event of failure such as connection timeout and SOAP fault, WSM will retry to resend the SOAP call back to the client. After failing to send the call back message 5 times in a row, the call back service will automatically use a 1 hour interval between each call. Normal operation will resume once WSM detects that it can again successfully send the message to the client SOAP server.

8 Test Environment

Vocus Wholesale provides a test environment in the form of a **sandbox**. The sandbox is a cut down version of WSM that is made available to aid in the development of the client application. The sandbox WSM interface is identical to the production system, but returns canned responses.

In the production environment requests undergo multiple stages of processing including authentication, verification, validation and then the actual processing by backend infrastructure. In contrast the sandbox will execute a SOAP request only up to the validation stage and then returns predefined responses. All asynchronous transactions result will be delivered via SOAP call-back.

Access to the testing environment is only available during the client's testing and development stage.

8.1 Accessing the Sandbox

To access the sandbox the client needs to configure the **operating system host file** with the following IP address:

Hostname	IP Address
wsm.webservice.m2.com.au	203.23.116.113

8.2 Simulating Successful Operation

To simulate a successful operation, send any **valid** request to the WSM. There is no concept of persistence in the sandbox, thus no data will be stored in database and all responses are predefined, including the TransactionID and *returned* ServiceID.

8.3 Simulating Failed Operation

To initiate any other error codes, the client can send the ProductID error code as the ServiceID or TransactionID of the SOAP request. These error codes must belong to the ProductID under testing.

For error code 4220 validation error, the client can send a request with **invalid** Params to the WSM. To simulate error code 4222 send an **invalid** ProductID to WSM.

Authentication, verification and flood protection errors can be initiated by sending an invalid certificate or flooding the server with requests. Asynchronous transaction result will be delivered via SOAP call-back.

8.4 Simulating WSM's Generated Transaction

To simulate WSM generated transaction, the Vocus Wholesale SOAP call-back client will call the client's SOAP server every 10 minutes with a canned SOAP call-back and notification request.